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Structural Selection in SO(10)-Motivated Unified Field Theories

A Phenomenological MAAT Layer for Gauge and Yukawa Benchmarks

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Abstract. Grand Unified Theories (GUTs) unify gauge interactions at the level of symmetry groups but leave the selection of physically realised vacuum states and parameter values largely unconstrained.
We introduce a *structural-selection* extension of SO(10) unified field theory within the MAAT framework. The construction defines a structural functional over admissible field configurations based on four primary sectors: equation consistency, conservation balance, dynamical activity, and inter-sector connectivity. Stability emerges as a derived robustness condition.
We extend this framework to parameter ranking by combining renormalization group evolution with a MAAT-weighted selection functional. Gauge and Yukawa parameters are treated as effective quantities whose admissible values can be selected or ranked by structural consistency, not as quantities derived from first principles in this first benchmark.
A two-parameter gauge benchmark demonstrates that structural selection identifies a viable GUT-scale region while failing precision unification at the one-loop level, indicating the necessity of threshold corrections, intermediate scales, or model-specific spectra. A third-generation Yukawa benchmark then shows that an SO(10)-motivated b–τ boundary condition can be made compatible with low-energy effective masses using a modest bottom-threshold correction. The framework provides a reproducible bridge between unified field dynamics, effective parameter selection, and structural selection.

§ 1 Introduction

Grand Unified Theories (GUTs), such as SO(10), provide a unified description of gauge interactions [1, 2, 3, 4]. However, they introduce a large degeneracy of admissible solutions, including multiple vacuum states and symmetry-breaking pathways. Standard formulations do not explain:

- ▷ why specific vacuum configurations are realised,
- ▷ why particular symmetry-breaking chains occur,
- ▷ why observed low-energy parameters take their values.

This suggests that an additional principle acts on the space of dynamically admissible configurations. We propose such a principle in the form of *structural selection*, implemented through the MAAT framework.

Status. This paper is a phenomenological bridge proposal. It does not construct a complete SO(10) model, prove gauge-coupling unification, derive the Standard Model spectrum from first principles, or solve the vacuum-selection problem. Its contribution is narrower: it defines a reproducible structural-selection layer over SO(10)-motivated gauge and Yukawa benchmarks and tests whether the resulting selection landscape is internally coherent.

§ 2 SO(10) Unified Field Theory

We consider a standard SO(10) gauge theory with action

$$S_{\mathrm{SO}(10)} = \int d^4x\sqrt{-g}\left[-\tfrac{1}{4}\,\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu}) + \bar{\Psi}_{16}\, i\gamma^\mu D_\mu\, \Psi_{16} + |D_\mu\Sigma|^2 - V(\Sigma) + \mathcal{L}_Y\right], \tag{1}$$

where Ψ_{16} contains one fermion generation including the right-handed neutrino, and Σ drives symmetry breaking.

§ 3 Structural Selection Functional

We define the MAAT structural functional

$$F_{\mathrm{MAAT}}[\Phi] = -\sum_{a\in\{H,B,S,V\}} \lambda_a \log\bigl(\epsilon + \Gamma_a[\Phi]\bigr) - \lambda_{\mathrm{cl}} \log\bigl(\epsilon + R_{\mathrm{rob}}[\Phi]\bigr), \tag{2}$$

inducing the selection measure

$$P[\Phi] \propto \exp\bigl(-F_{\mathrm{MAAT}}[\Phi]\bigr). \tag{3}$$

§ 4 Sector Definitions

The four primary structural sectors are defined by the deviations

$$d_H \sim \sum_i \left|\frac{\delta S}{\delta \Phi_i}\right|^2, \tag{4}$$

$$d_B \sim \left|\nabla_\mu T^{\mu\nu}\right|^2 + |D_\mu J^\mu|^2, \tag{5}$$

$$d_S \sim \left(|D_\mu \Sigma|^2 - \chi\right)^2, \tag{6}$$

$$d_V \sim 1 - \tilde{I}_{\text{sector}}, \tag{7}$$

with structural supports $\Gamma_a = (1 + d_a)^{-1} \in (0, 1]$. Here $\tilde{I}_{\text{sector}} \in [0, 1]$ denotes a normalised sectoral mutual-information or correlation proxy. This normalisation is required so that $d_V \geq 0$.

§ 5 Emergent Robustness

Stability is not imposed externally but emerges as a derived robustness condition:

$$R_{\text{resp}} = (H \cdot B \cdot V)^{1/3}, \tag{8}$$

$$R_{\text{rob}} = \min\Big[(HBV)^{1/3},\ (HBSV)^{1/4}\Big]. \tag{9}$$

§ 6 Parameter Selection: Master Equation

We extend the framework to the gauge-coupling sector. The one-loop running is governed by

$$\alpha_i^{-1}(M_Z) = \alpha_{\text{GUT}}^{-1} + \frac{b_i}{2\pi} \ln\left(\frac{M_{\text{GUT}}}{M_Z}\right) + \Delta_i, \tag{10}$$

and the total selection functional reads

Total Selection Functional

$$\mathcal{L}_{\text{total}} = \chi^2_{\text{SM}} + F_{\text{MAAT}} + F_{\text{threshold}},$$

with

$$\chi^2_{\text{SM}} = \sum_i \left(\frac{\alpha_i^{\text{pred}} - \alpha_i^{\text{obs}}}{\sigma_i}\right)^2, \tag{11}$$

$$F_{\text{threshold}} = \sum_i \left(\frac{\Delta_i}{\sigma_\Delta}\right)^2. \tag{12}$$

The optimal parameters are determined by

Minimisation Principle

$$\left(\alpha_{\mathrm{GUT}},\, M_{\mathrm{GUT}},\, \Delta_i\right) \;=\; \arg\min\, \mathcal{L}_{\mathrm{total}}.$$

§ 7 Numerical Benchmark: Gauge Sector

A two-parameter fit without threshold corrections yields:

Quantity	Value	Note
α_{GUT}	0.0227	GUT coupling
M_{GUT}	6.6×10^{15} GeV	unification scale
χ^2	100.8	one-loop residual
H	0.996	structural support
R_{rob}	0.60	robustness

The result indicates a structurally stable GUT region with insufficient precision at one-loop level, motivating threshold corrections or intermediate scales.

§ 8 Physical Interpretation

The MAAT selection functional introduces a conceptual shift:

Gauge couplings are effective parameters whose admissible values can be ranked by structural consistency.

This extends naturally to the Yukawa sector. The Yukawa selection term is

$$F_{\mathrm{Yukawa}} = -\sum_f \lambda_f \log\left(\epsilon + \Gamma_f\right), \tag{13}$$

where Γ_f encodes consistency of renormalization-group flow, hierarchical structure, and flavor mixing. In the present benchmark, fermion masses are not derived from first principles; rather, the low-energy effective Yukawa values are tested for compatibility with an SO(10)-motivated high-scale boundary condition.

§ 9 Yukawa Sector and High-Scale Unification

We consider the SO(10)-motivated third-generation boundary condition

$$y_b(M_{\mathrm{GUT}}) = y_\tau(M_{\mathrm{GUT}}), \tag{14}$$

with one-loop renormalization group evolution

$$16\pi^2 \frac{dy_f}{d\ln\mu} = \beta_f\big(y_t, y_b, y_\tau, g_1, g_2, g_3\big), \qquad f \in \{t, b, \tau\}, \tag{15}$$

and a low-energy bottom threshold correction

$$y_b^{\text{eff}}(M_Z) = y_b^{\text{RG}}(M_Z)\,(1 + \Delta_b). \tag{16}$$

The total selection functional in the Yukawa sector is

$$\mathcal{L}_{\text{total}} = \chi^2_{\text{Yukawa}} + F_{\text{MAAT}} + \left(\frac{\Delta_b}{\sigma_{\Delta_b}}\right)^2 + F_{\text{complexity}}. \tag{17}$$

9.1 Numerical Result: Coupled RG Benchmark (v0.4)

Quantity	**Value**	**Note**
M_{GUT}	1.86×10^{16} GeV	
$y_t(M_{\text{GUT}})$	0.414	top Yukawa
$y_b = y_\tau$ at M_{GUT}	0.00774	b-τ unification
Δ_b	0.051	∼5% threshold
m_t^{pred}	172.76 GeV	
m_b^{pred}	4.18 GeV	
m_τ^{pred}	1.78 GeV	
$y_b/y_\tau\,(M_Z)$	2.35	consistent with effective input

The structural supports are near unity:

$$H \approx 1, \quad B \approx 1, \quad S \approx 1, \quad V \approx 0.998, \quad R_{\text{rob}} \approx 0.999. \tag{18}$$

This indicates that the solution is not only numerically accurate but also structurally coherent within the MAAT framework — the structural and physical consistency conditions are simultaneously satisfied.

On the role of structural parameters. At first sight, the introduction of sector weights λ_a may appear to shift the parameter problem rather than resolve it. However, in the MAAT framework these coefficients are not treated as free parameters.

Instead, they are determined dynamically through a response-closure relation of the form

$$\lambda = \left(C + \eta\,\frac{\text{tr}(C)}{A} I\right)^{-1} (\langle d\rangle - d^*), \tag{19}$$

where C denotes the defect covariance matrix.

This construction implies that the weights are emergent quantities derived from the statistical structure of the theory, rather than externally imposed inputs.

Quantum corrections and stability. The present analysis is performed at the one-loop level and does not include higher-order quantum corrections. However, the MAAT functional can be naturally embedded into a path-integral formulation,

$$Z = \int \mathcal{D}\Phi \, \exp\left(iS_0[\Phi] - F_{\mathrm{MAAT}}[\Phi]\right), \tag{20}$$

where S_0 denotes the underlying quantum field theory action.

In this formulation, the MAAT contribution acts as a non-perturbative weighting factor over configurations. This suggests that structural selection may remain meaningful beyond the semiclassical level, although a full quantum treatment is left for future work.

Definition of the connectivity sector. The connectivity sector V is defined in terms of correlations between field components. In practical implementations, we use a mutual information measure,

$$d_V = 1 - \tilde{I}_{\mathrm{nn}}, \tag{21}$$

where $\tilde{I}_{\mathrm{nn}} \in [0, 1]$ denotes the normalised nearest-neighbour mutual information between field degrees of freedom.

This provides a quantitative measure of inter-sector coherence, ensuring that dynamically coupled fields remain structurally correlated rather than decoupling into independent subsystems.

§ 10 Formal Structure of the MAAT Functional

10.1 Structural Functional and Selection Measure

The MAAT framework introduces a structural functional over field configurations Φ:

$$F_{\mathrm{MAAT}}[\Phi] = -\sum_{a\in\{H,B,S,V\}} \lambda_a \log(\epsilon + \Gamma_a[\Phi]) - \lambda_{\mathrm{cl}} \log(\epsilon + R_{\mathrm{rob}}[\Phi]). \tag{22}$$

The corresponding selection measure is defined as

$$P[\Phi] \propto \exp(-F_{\mathrm{MAAT}}[\Phi]). \tag{23}$$

This induces a probabilistic weighting over dynamically admissible configurations.

10.2 Support Fields and Defects

Each structural sector is defined through a non-negative defect $d_a[\Phi]$:

$$\Gamma_a[\Phi] = \frac{1}{1 + d_a[\Phi]}, \qquad d_a \geq 0. \tag{24}$$

Harmony (H): Equation Consistency

$$d_H = \sum_i \left| \frac{\delta S}{\delta \Phi_i} \right|^2 . \tag{25}$$

Balance (B): Conservation and Constraints

$$d_B = |\nabla_\mu T^{\mu\nu}|^2 + |D_\mu J^\mu|^2. \tag{26}$$

Activity (S): Controlled Dynamics

$$d_S = \left(|D_\mu \Sigma|^2 - \chi \right)^2 . \tag{27}$$

Connectivity (V): Correlation Structure The connectivity sector is defined via mutual information between field degrees of freedom:

$$d_V = 1 - \tilde{I}_{\mathrm{nn}}, \tag{28}$$

where $\tilde{I}_{\mathrm{nn}} \in [0, 1]$ denotes the normalised nearest-neighbour mutual information. Alternatively, in a field-theoretic representation:

$$d_V = \sum_{i<j} \left| \nabla_\mu \Phi_i \nabla^\mu \Phi_j - \langle \nabla \Phi_i \cdot \nabla \Phi_j \rangle \right| . \tag{29}$$

10.3 Emergent Robustness Closure

Robustness is not an independent sector but emerges from the primary supports:

$$R_{\mathrm{resp}} = (HBV)^{1/3}, \tag{30}$$

$$R_{\mathrm{rob}} = \min \left[(HBV)^{1/3}, (HBSV)^{1/4} \right] . \tag{31}$$

This defines the stability of the configuration:

$$\text{Stability} = R_{\mathrm{rob}}. \tag{32}$$

§ 11 Response Closure of Structural Weights

The sector weights λ_a are not treated as free parameters. Instead, they are determined through a response-closure relation:

$$\lambda = \left(C + \eta \frac{\mathrm{tr}(C)}{A} I \right)^{-1} (\langle d \rangle - d^*), \tag{33}$$

where:

- ▷ C is the covariance matrix of defects,
- ▷ $\langle d \rangle$ denotes ensemble averages,

- d^* is a target defect vector,
- η is a regularization parameter.

This ensures that the structural weights are emergent quantities derived from the statistical structure of the theory.

§ 12 Effective Action and Path Integral Formulation

The MAAT functional can be incorporated into the effective action:

$$S_{\text{eff}} = S_0 + F_{\text{MAAT}}. \tag{34}$$

This leads to a modified path integral:

$$Z = \int \mathcal{D}\Phi \, \exp\left(iS_0[\Phi] - F_{\text{MAAT}}[\Phi]\right). \tag{35}$$

In this formulation, the MAAT term acts as a non-perturbative weighting factor over field configurations.

§ 13 Parameter Selection Framework

13.1 Gauge Sector

Gauge couplings are determined by minimizing:

$$\mathcal{L}_{\text{gauge}} = \chi^2_{\text{SM}} + F_{\text{MAAT}} + F_{\text{threshold}}. \tag{36}$$

13.2 Yukawa Sector

The Yukawa sector is incorporated through coupled RG evolution:

$$16\pi^2 \frac{dy_f}{d\ln\mu} = \beta_f(y_t, y_b, y_\tau, g_1, g_2, g_3), \tag{37}$$

with high-scale boundary condition:

$$y_b(M_{\text{GUT}}) = y_\tau(M_{\text{GUT}}). \tag{38}$$

Low-energy threshold corrections are included via:

$$y_b^{\text{eff}}(M_Z) = y_b^{\text{RG}}(M_Z)(1 + \Delta_b). \tag{39}$$

The total selection functional becomes:

$$\mathcal{L}_{\text{total}} = \chi^2_{\text{Yukawa}} + F_{\text{MAAT}} + \left(\frac{\Delta_b}{\sigma_{\Delta_b}}\right)^2 + F_{\text{complexity}}. \tag{40}$$

§ 14 Numerical Implementation

The parameter optimization is performed using a hybrid approach:

- Global optimization via differential evolution,
- Local refinement via bounded gradient-based methods.

Constraints are enforced explicitly to ensure physical consistency:

$$\alpha_{\mathrm{GUT}} > 0, \quad M_{\mathrm{GUT}} > M_Z, \quad y_f > 0. \tag{41}$$

§ 15 Interpretation

The MAAT framework replaces unconstrained parameter insertion with an explicit selection problem. The parameters remain effective model parameters, but their preferred regions are ranked by structural consistency rather than chosen only by a low-energy fit.

The combined system

$$\Phi^*, \theta^* = \arg\min \left(S_0[\Phi] + F_{\mathrm{MAAT}}[\Phi] + \mathcal{L}_{\mathrm{obs}}(\theta) \right) \tag{42}$$

defines both the physical field configuration Φ^* and the effective parameters θ^* as outcomes of a unified selection principle.

The selection landscape exhibits a pronounced valley structure, indicating that the MAAT functional does not select a single isolated point but rather a narrow manifold of structurally consistent solutions.

The global minimum is located near

$$M_{\mathrm{GUT}} \sim 10^{16.2}\ \mathrm{GeV}, \quad y_{b\tau} \sim 7.7 \times 10^{-3}.$$

This behaviour suggests that parameter determination occurs through a constrained selection geometry rather than a point-like fine-tuning.

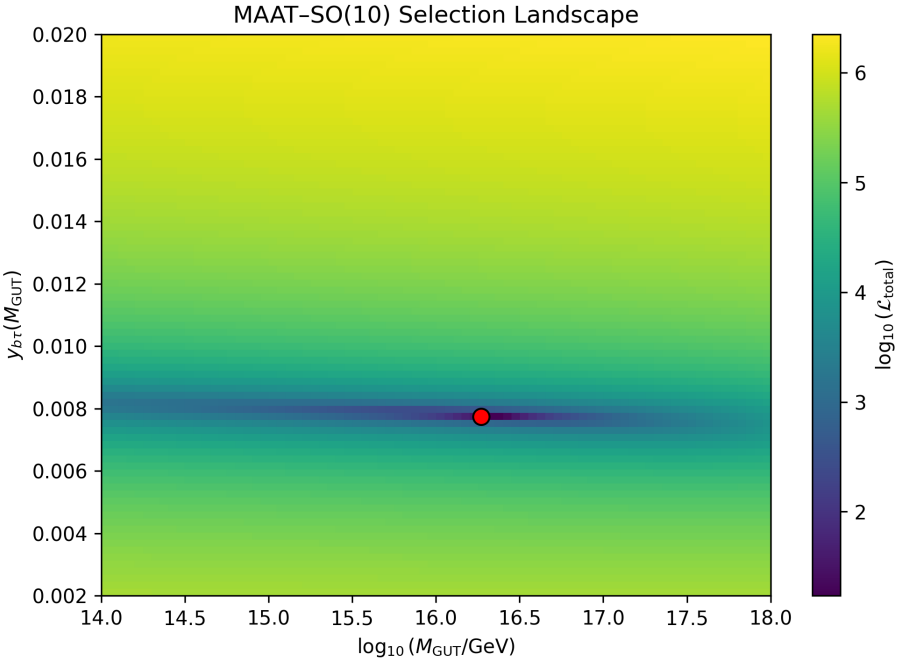

Figure 1: MAAT–SO(10) Yukawa selection landscape in the $(\log_{10} M_{\rm GUT}, y_{b\tau})$ plane. The red marker denotes the best-fit benchmark. The valley structure indicates a constrained structural region rather than an isolated point prediction.

§ 16 Reproducibility and Data Attribution

The associated repository is:

https://github.com/Chris4081/structural-selection-principle

The relevant experiment folder is:

`experiments/maat_so10_structural_selection/`

The benchmark scripts are:

```
python3 maat_so10_gauge_2d_fit.py
python3 maat_so10_yukawa_coupled_rg_v04.py
```

The numerical inputs use standard electroweak-scale values: M_Z, approximate gauge couplings, and third-generation fermion masses. These are taken from the Particle Data Group review values and used here as effective low-energy inputs [5]. For this first benchmark we use effective Yukawa couplings derived from fermion masses at the electroweak scale, without performing a full scheme conversion between pole and $\overline{\rm MS}$ definitions.

Data and licence note. PDG values are external scientific reference data and should be cited to the Particle Data Group when reused or discussed. Repository CSV/PNG/PDF artifacts are derived analysis and visualisation outputs only. No endorsement by the Particle Data Group or by any external collaboration is implied.

§ 17 Limitations

The current analysis does not yet include:

- ▷ a complete SO(10) Higgs-sector construction or explicit breaking chain,
- ▷ two-loop renormalization group corrections,
- ▷ threshold-resolved GUT spectra,
- ▷ realistic neutrino-sector fitting,
- ▷ proton-decay constraints,
- ▷ a full conversion between pole and $\overline{\mathrm{MS}}$ mass schemes,
- ▷ a full quantum treatment of the structural functional.

The gauge-sector result should therefore be read as a one-loop diagnostic: it locates a plausible GUT-scale region but does not achieve precision unification. The Yukawa-sector result should be read as an SO(10)-motivated compatibility benchmark with a fitted bottom-threshold correction, not as a first-principles derivation of fermion masses.

§ 18 Conclusion

We have introduced a structural-selection extension of SO(10) grand unified theories that elevates parameter determination from a fine-tuning problem to a *selection problem*.

The MAAT functional acts as a structural free energy over the joint space of field configurations and parameters, providing an effective framework for vacuum ranking and parameter selection. In this benchmark, gauge couplings and Yukawa couplings are not derived from first principles; instead, their admissible regions are ranked by structural consistency in a way that is reproducible and falsifiable by more complete GUT calculations.

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